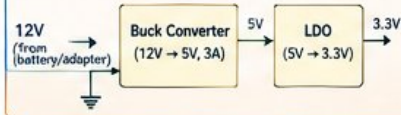
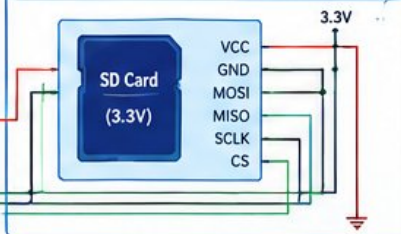
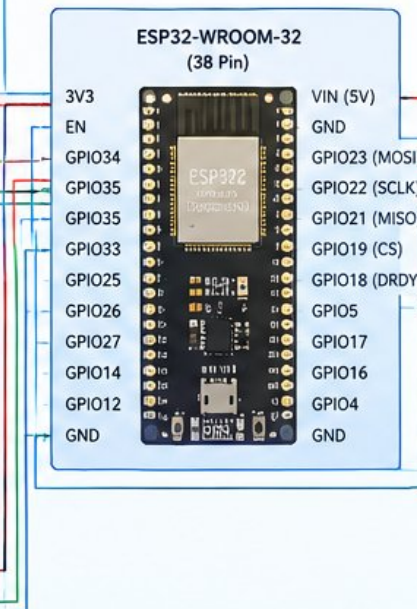
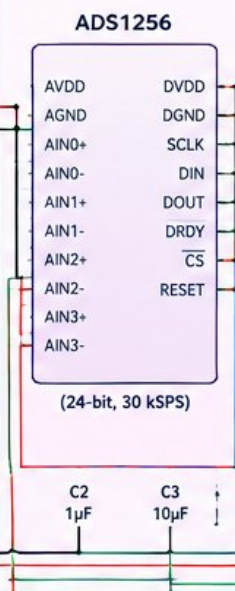
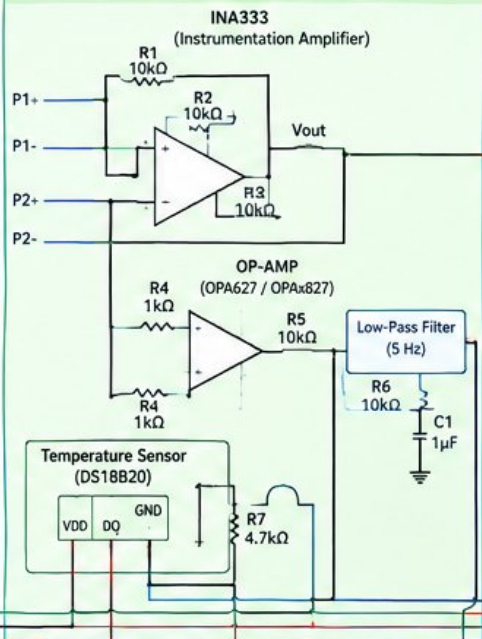
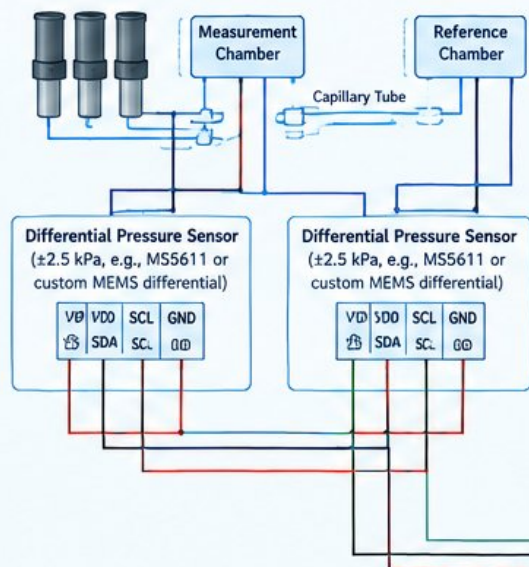
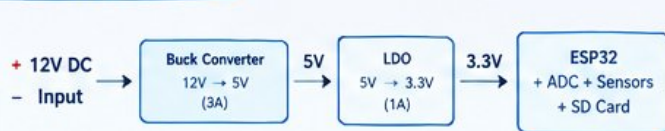


Target Frequency Range : 0.01 – 20 Hz
Resolution : 24-bit | Sampling : 50 – 200 Hz



Function	ESP32 Pin
SCLK (ADC & SD)	GPIO22
MOSI (ADC & SD)	GPIO23
MISO (ADC & SD)	GPIO19
CS (ADC)	GPIO19
DRDY (ADC)	GPIO18
SD Card CS	GPIO5
Wi-Fi	Built-in
I2C (Pressure Sensors)	GPIO21 (SDA) / GPIO22 (SCL)
Temperature Sensor	GPIO4 (DQ)

No.	Component	Qty
1	ESP32-WROOM-32	1
2	ADS1256	1
3	INA333	1
4	OPA627 / OPAx827	1
5	MS5611 (or similar)	2
6	DS18B20	1
7	SD Card Module	1
8	12V to 5V Buck Converter	1
9	5V to 3.3V LDO	1
10	Passive Components	1 set



- Use 3.3V logic levels for ESP32, SD card and ADC.
- Keep **analog** and **digital** grounds separate (star connection).
- Use low-noise, precision op-amps (OPA627 + OPAx827).
- Add proper decoupling capacitors ($0.1\mu\text{F} + 10\mu\text{F}$) near ICs.
- Use shielded cables for sensor lines to reduce noise.
- Ensure good PCB layout with short traces for analog signals.
- Power supply should be stable and low-noise.



NTRO – Smart India Hackathon

High-Sensitivity Microbarometer / Infrasound Sensor – Prototype Circuit & Signal Chain

